

DS&AI

Artificial Intelligence

DPP 02

Un- Informed Search

Q1 Underlying each search problem is a search graph in which the states are the vertices of the search graph and the edges (connections) are determined by the possible actions; there is an edge from s to s' with label a in the search graph, if and only if there is an action a which leads from state s to state s' . These two conditions together completely define the search graph. A cycle in the search graph is a vertex (state) s and a non-empty sequence of actions as such that if we start in s and execute the sequence of actions as then we will end up in s again. Consider the following statements about search problems and graphs: graphs:

- If the Breadth-First Search Graph Algorithm does terminate on a given search problem then the corresponding search graph contains no cycles.
- If the corresponding search graph is finite and contains no cycles then the Depth First Search Tree Algorithm always terminates on the corresponding search problem.

Which of the following claims is true?

- Both statements (i) and (ii) are false.
- Only statement (i) is true.
- Both statements (i) and (ii) are true.
- Only statement (ii) is true.

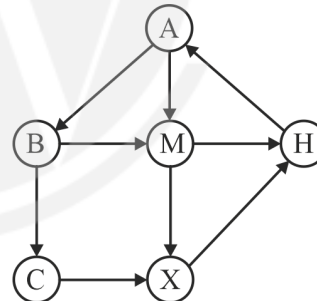
Q2 Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered $n + 1$ and $n + 2$. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list. Which ONE of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?

- BFS expands more states than DFS.
- DFS expands more states than BFS.
- Both BFS and DFS expand equal number of states.
- Both BFS and DFS do not reach the goal state number 6.

Q3 Let b be the branching factor of a search tree. If the optimal goal is reached after d actions from the initial state, in the worst case, how many times will the initial state be expanded for iterative deepening depth-first search (IDDFS) and iterative Deepening A* search (IDA*)?

- IDDFS- d , IDA*- d .
- IDDFS - d , IDA - b^d
- IDDFS - b^d , IDA*- d .
- IDDFS - b^d , IDA*- b^d .

Q4 Consider the following directed graph.



There are several possible DFS orderings starting from node A. Consider the following DFS ordering:

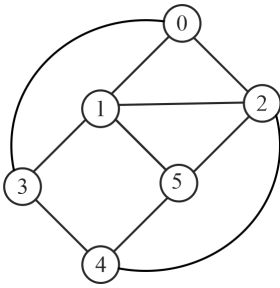
A, B, C, X, H, M.

Which of the following statements is/are TRUE for the given DFS ordering?

- The edge (M, X) is a cross edge.
- The edge (M, H) is a cross edge.
- The edge (H, A) is a back edge.
- The edge (B, M) is a cross edge.

Q5 Which sequence corresponds to that of depth first search for the graph given below.

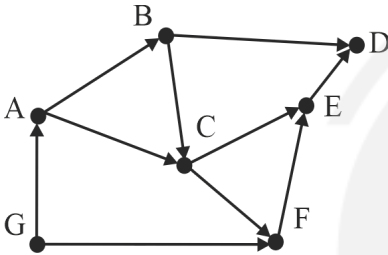




The search starts at vertex 0 and lexicographic ordering is assumed for the edges emanating from each vertex.

- (A) 012435 (B) 012543
(C) 012345 (D) 013425

- Q6** For the graph shown in Figure 1, suppose vertices A, B, C, D, E, F, G are stored in alphabetical order in this graph. What is DFS order of traversing this graph starting from vertex G?



- (A) G, A, B, C, D, E, F
(B) G, A, B, D, C, E, F
(C) G, A, B, C, E, D, F
(D) G, A, B, C, F, E, D

- Q7** Which of the following statements about Depth-Limited Search is FALSE?
- (A) It uses less memory compared to depth-first search.
(B) It is guaranteed to find a solution if the depth limit is set correctly.
(C) It can always guarantee optimality if the depth limit is sufficient.
(D) It can be incomplete if the depth limit is too low.

- Q8** Consider a state space where the start state is number 1. The successor function for the state numbered n returns two states numbered $n+1$ and $n+2$. Assume that the states in the unexpanded state list are expanded in the ascending order of numbers and the previously expanded states are not added to the unexpanded state list.

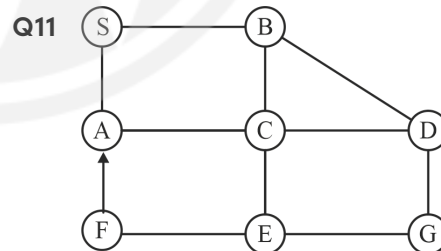
Which ONE of the following statements about breadth-first search (BFS) and depth-first search (DFS) is true, when reaching the goal state number 6?

- (A) PFS expands more states than DFS.
(B) DFS expands more states than BFS.
(C) Both BFS and DFS expand equal number of states.
(D) Both BFS and DFS do not reach the goal state number 6.

- Q9** Which of the following statements accurately describes the concept of state space?
- (A) State space is the set of all actions that an agent can take in a given environment.
(B) State space is a representation of an agent's internal thought processes.
(C) State space is the set of all states that can be reached from the initial state by any sequence of actions.
(D) State space is a measure of an agent's computational complexity.

- Q10** Consider the state space with the branching factor of 4 and assume that goal occurs somewhere at depth 6.

How many maximum numbers of nodes will be inspected if we used BFS search algorithm?



Note that MoveGen determines the order in which neighbors are generated.

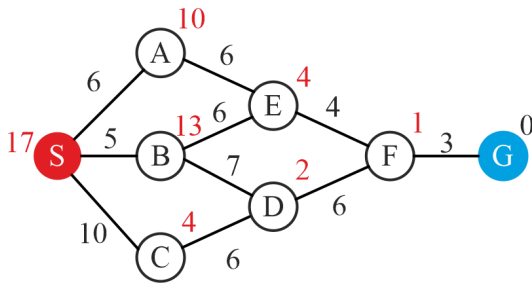
X	MoveGen(x)
S	[A,B]
A	[C, S]
B	[C,D,S]
C	[E,A,D,B]
D	[G,C,B]
E	[F, C, G]
F	[A, E]
G	[D, E]



At the instance when DFS finds G, the OPEN list is _____.

- (A) B,D,G (B) D,G
(C) F,G (D) E,G

Q12

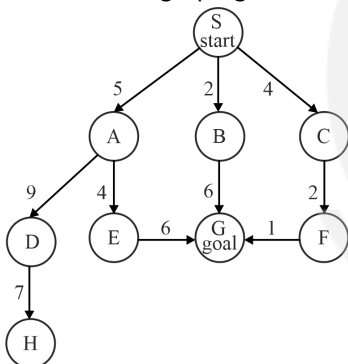


Here S is the start node and G is the goal node.

Note - Use alphabetical order for tie-breaker. The edge cost is given on the edge and the heuristic value for each node is given on the node circle. Assume nodes are generated in alphabetical order.

Apply uniform cost search algorithm, to find out the shortest path and cost of shortest path.

Q13 Consider the graph given below



If we apply Uniform Cost Search Algorithm, then what will be the path to reach node G from node S?

- (A) $S \rightarrow C \rightarrow F \rightarrow G$
(B) $S \rightarrow B \rightarrow G$
(C) $S \rightarrow A \rightarrow E \rightarrow G$
(D) $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Q14 What will be the max number of nodes generated in the Iterative deepening search with dept $d = 5$ and branching factor = 10?

Q15 Consider the Edges (E) and Vertices (V) of the graph G as shown below

$$V(G) = \{a, b, c, d, e, f, g\}$$

$$E(G) = \{ab, bc, bd, de, ef, fc, fg\}$$

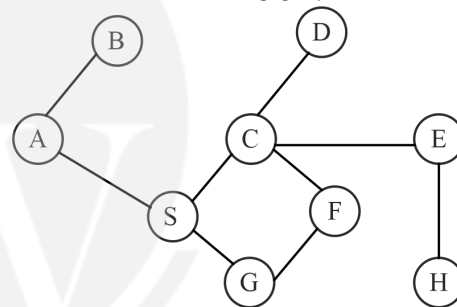
Let G_1 be the graph obtained after applying the BFS to the graph G. Which of the following will represent the Edges of graph G_1 i.e. $E(G_1)$?

- (A) $OE(G_1) = \{ab, bd, de, bc, fg\}$
(B) $E(G_1) = \{ab, bd, ef, de, fg, fc\}$
(C) $OE(G_1) = \{ab, bd, bc, de, cf, fg\}$
(D) $E(G_1) = \{ab, bd, de, ef, fg, fc\}$

Q16 Consider a graph where each edge has a different cost. If you are using the Uniform Cost Search algorithm to find the shortest path from a start node A to a goal node G, which of the following statements is TRUE regarding the characteristics of UCS?

- (A) UCS is complete, but not optimal
(B) UCS is both complete and optimal.
(C) UCS can only be used in graphs with uniform edge costs.
(D) UCS uses a depth-first strategy

Q17 Consider the following graph:



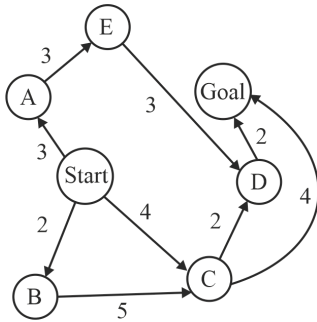
What is the correct DFS traversal starting from node?

Assume that you visit alphabetically smaller nodes first.

- (A) ABSCDEHGF (B) ABSCDEGFH
(C) ABSCHEDGF (D) ABCSDEHGF

Q18 In the below directed graph, what will be the nodes that still remain in OPEN list when "Goal nodes is reached by DFS (depth first search) algorithm?

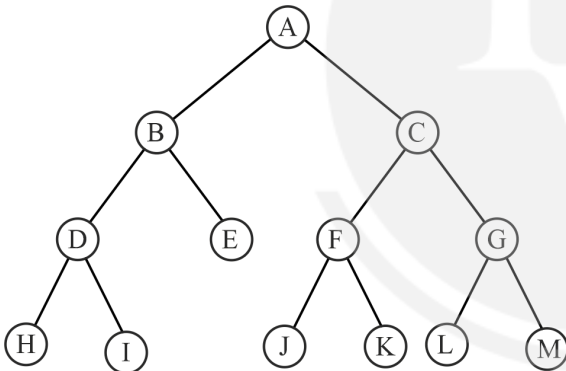




Note - When two nodes have same cost then take node in alphabetical order for tie breaker. And also function generates neighbour alphabetically.

- (A) E,D (B) B,C
(C) AB (D) A,B,D

- Q19** Consider the following graph in which we are searching from start state A to the goal state J. What is the number of nodes generated by Iterative Deepening Search? Assume that nodes are explored in right-to-left order and all edge costs are 1. Note: Exploration of the goal state J should be added to the count.



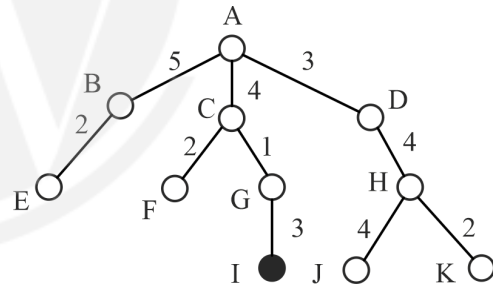
- Q20** Consider the depth-limited search algorithm with depth limit f . If the only goal state exists at depth d then which of the following statement(s) is/are correct?
- (A) If $f > d$ then it is complete but not optimal.
(B) If $f > d$ then it is not complete and not optimal.
(C) If $f < d$ then it is complete but not optimal.
(D) If $f < d$ then it is not complete and not optimal.
- Q21** Consider the following terminology and match List I with List II and choose the correct answer from the code given below.
- b = branching factor

d = depth of the shallowest solution
 m = maximum depth of the search tree
 l = depth limit

List-I	(Algorithms)	List II	(Space Complexity)
(a)	BFS search	(i)	$O(bd)$
(b)	DFS search	(ii)	$O(bd)$
(c)	Depth-limited search	(iii)	$O(bm)$
(d)	Iterative deepening search	(iv)	$O(bl)$

- (A) (a) (i), (b)-(ii), (c)-(iv), (d)-(iii)
(B) (a) (ii), (b)-(iii), (c)-(iv), (d)-(i)
(C) (a) (iii), (b)-(ii), (c)-(iv), (d)-(i)
(D) (a) (i), (b)-(iii), (c)-(iv), (d)-(ii)

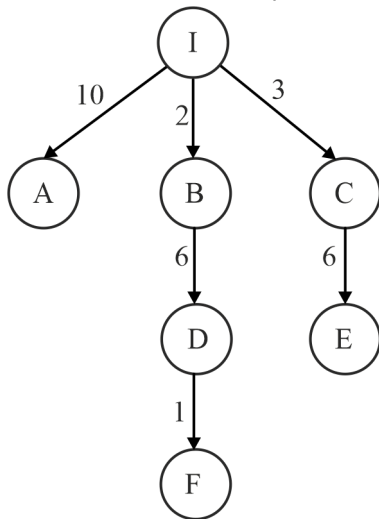
- Q22** For the following tree, show the order of nodes visited for breadth-first search, depth-first search, uniform cost search, and iterative deepening search. The goal node is I and the numbers next to the edges indicate the associated cost



- Q23** You are running UCS in the state space graph below. You just expanded (visited) node C. What

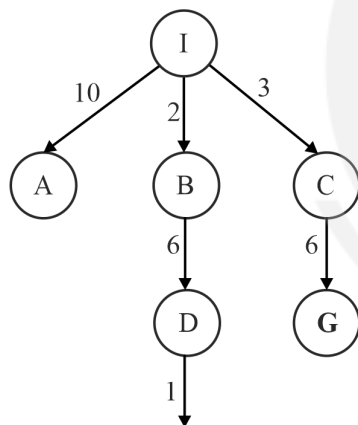


node will the search expand next?



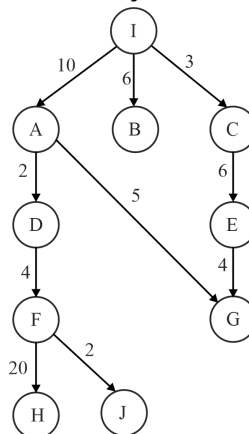
- (A) A (B) D
(C) E (D) F

Q24 You are running DFS in the state space graph below. DFS expands nodes left to right. G is the goal state. The state space graph is infinite (the path after D does not terminate). What is the behavior of DFS?



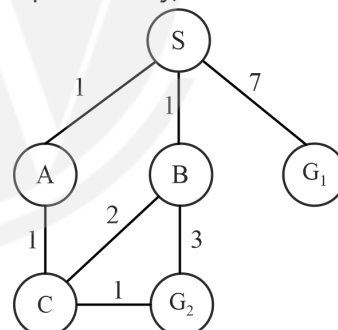
- (A) Get stuck in an infinite loop
(B) Return A
(C) Return G
(D) Return "failure"

Q25 Consider the state space graph below. Goal states borders. Nodes are expanded left to right when there are ties. What solution path is returned by UCS ?



- (A) IADFH (B) IADFJ
(C) IAG (D) ICEG

Q26 Which goal is reached and what is the total cost of the solution found for the following state-space graph when using Breadth-First Search and Uniform-Cost Search (S is the start state, G1 and G2 are the goal states, arcs are bidirectional, no repeated state checking, break any ties alphabetically)?



- (A) BFS: G1 (Cost: 7), UCS: G2 (Cost: 4)
(B) BFS: G2 (Cost: 4), UCS: G1 (Cost: 7)
(C) BFS: G2 (Cost: 4), UCS: G2 (Cost: 4)
(D) BFS: G1 (Cost: 7), UCS: G2 (Cost: 3)



Answer Key

Q1 (D)
Q2 (C)
Q3 (A)
Q4 (A)
Q5 (A)
Q6 (C)
Q7 (C)
Q8 (C)
Q9 (C)
Q10 546~546
Q11 (D)
Q12 18~18
Q13 (A)

Q14 123456
Q15 (C)
Q16 (B)
Q17 (A)
Q18 (A)
Q19 19~19
Q20 (A)
Q21 (B)
Q22 9~9
Q23 (B)
Q24 (A)
Q25 (D)
Q26 (D)



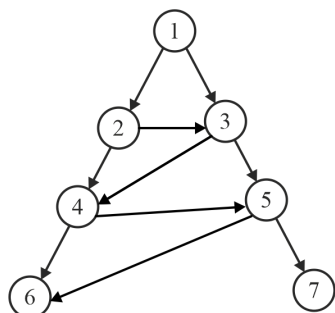
Hints & Solutions

Q1 Text Solution:

Only statement (ii) is true.

Q2 Text Solution:

Both BFS and DFS expand equal number of states.



BFS close 1 2 3 4 5 6

Open 1 2 3 4 5 6 7

DFS close 1 2 3 4 5 6

Open
1 3 2 4 3 5 4 6 5 7 6

Q3 Text Solution:

- IDDFS we need to do max no. of iteration
⇒ depth of tree = d
- IDA* (b^d) times = max, number of iteration =
max number of node b^d

Q4 Text Solution:

Close 0 1 2 3 4 5

Open
0 3 2 1 3 5 2 5 4 5 3

Q5 Text Solution:

Close 0 1 2 3 4 5

Open
0 3 2 1 3 5 2 5 4 5 3

Q6 Text Solution:

Close: G A B C E D F

Open
G F A C B D E F D

Q7 Text Solution:

- It can always guarantee optimality if the depth limit is sufficient.

Q8 Text Solution:

- Both BFS and DFS expand equal number of states.

Q9 Text Solution:

- State space is the set of all states that can be reached from the initial state by any sequence of actions.

Q10 Text Solution:

$$(4^0 + 4^1 + 4^2 + 4^3 + 4^4 + 4^5 + 4^6) = \frac{4^7 - 1}{3} = 546$$

Q11 Text Solution:

Close: S A C B D G

Open
~~S~~ B ~~A~~ ~~C~~ E D ~~B~~ ~~D~~ ~~G~~

Q12 Text Solution:

	S	B	A	C	E	D	F	G
S	0	0	0	0	0	0	0	0
A	6	6	6	6	6	6	6	6
B	5	5	5	5	5	5	5	5
C	10	10	10	10	10	10	10	10
E	-	11	11	11	11	11	11	11
D	-	12	12	12	12	12	12	12
F	-	-	-	-	15	15	15	15
G	-	-	-	-	-	-	-	18

$S \rightarrow D \rightarrow B \rightarrow E \rightarrow F \rightarrow G = 18$

Q13 Text Solution:

$S \rightarrow C \rightarrow F \rightarrow G$

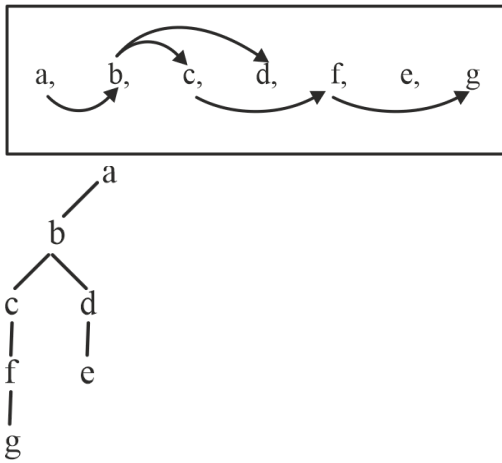
Q14 Text Solution:

- $10^0 \rightarrow 1$
 - $10^0 + 10^1 \rightarrow 11$
 - $10^0 + 10^1 + 10^2 \rightarrow 111$
 - $10^0 + 10^1 + 10^2 + 10^3 \rightarrow 1111$
 - $10^0 + 10^1 + 10^2 + 10^3 + 10^4 \rightarrow 11111$
 - $10^0 + 10^1 + 10^2 + 10^3 + 10^4 + 10^5 \rightarrow 111111$
- add = 123456

Q15 Text Solution:

BFS

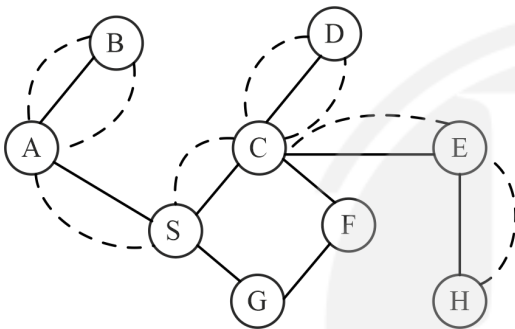




Q16 Text Solution:

- UCS is both complete and optimal.

Q17 Text Solution:



Q18 Text Solution:

S A E D G

~~X~~ C B ~~X~~ ~~X~~ ~~X~~ G

Q19 Text Solution:

A

ABC

ABCDEFG

ACGMLEKJ

Q20 Text Solution:

- If $f > d$ then it is complete but not optimal.

Q21 Text Solution:

List-I (Algorithms)	List-II (Space Complexity)
(a) BFS search	(i) $O(bd)$
(b) DFS search	(ii) $O(bd)$
(c) Depth-limited search	(iii) $O(bm)$
(d) Iterative deepening search	(iv) $O(bl)$

Q22 Text Solution:

- BDF ABCDEFGHI
- DFS ABE CFGI
- IDDFS A
A B C D
A B E C F G D H

A D E C F G I

UCS

	A	D	C	B	G	F	H	E	I
A	0	0	0	0	0				
B	5	5	5	5	5				
C	4	4	4	4	4				
D	3	3	3	3	3				
H	×	7	7	7	7				
F	×	×	6	6	6				
G	×	×	5	5	5				
E	×	×	×	7	7				
I	×	×	×	×	8				
J	×	×	×	×	×	×	4		
K	×	×	×	×	×	×	9		

Q23 Text Solution:

I	I	B	C	D
A	10	10	10	
B	2	2	2	
C	3	3	3	
D	×	8	8	
E	×	×	9	

Q24 Text Solution:

IA + B.D Infinite

Q25 Text Solution:

I	I	C	B	E	A	D	G
A	10	10	10	10	10	10	
B	6	6	6	6	6	6	
C	3	3	3	3	3	3	
E	×	9	9	9	9	9	
G	×	×	×	13	13	13	
D	×	×	×	×	12	12	
F	×	×	×	×	×	16	

Q26 Text Solution:

BFS: G1 (Cost: 7), UCS: G2 (Cost: 3)

